

Linux Incident Response & Threat Hunting Investigation

This project demonstrates a Linux-focused incident response and threat hunting investigation using Auditd forensic logs. The investigation simulates a Security Operations Center (SOC) workflow by identifying attacker behavior, tracing malicious processes, detecting privilege escalation, and investigating persistence mechanisms within a compromised Linux environment.

The project focuses on real-world blue team investigation techniques commonly used by SOC analysts, incident responders, and threat hunters.

Project Objectives

- Investigate suspicious Linux Auditd activity
- Detect and analyze reverse shell behavior
- Trace malicious process execution using PID correlation
- Investigate privilege escalation activity
- Identify persistence mechanisms
- Correlate suspicious outbound network communication
- Map attacker behavior to the MITRE ATT&CK Framework
- Produce a professional incident investigation report

Skills Demonstrated

- Linux Threat Hunting
- Incident Response
- Process Tree Analysis
- PID Correlation
- Privilege Escalation Detection
- Persistence Detection
- Auditd Log Analysis
- Command-Line Forensics
- MITRE ATT&CK Mapping

- Security Documentation & Reporting

Tools Used

- Linux Auditd
- ausearch
- grep
- Linux Shell
- Process Correlation Techniques
- MITRE ATT&CK Framework

Investigation Workflow

1. Environment Preparation

- Accessed Linux investigation environment
- Enabled root privileges
- Validated Auditd log availability

2. Reverse Shell Detection

- Identified suspicious Socat reverse shell activity
- Detected outbound TCP communication
- Investigated attacker shell execution

3. Process Tracing

- Traced malicious process chain using PIDs
- Identified parent-child process relationships
- Determined original malicious source process

4. Privilege Escalation Investigation

- Detected su root execution
- Confirmed successful root privilege escalation

- Investigated post-escalation shell activity

5. Persistence Investigation

- Detected cron-based persistence
- Investigated SSH authorized_keys modification
- Identified suspicious systemd service creation

6. Threat Hunting & Correlation

- Correlated shell execution behavior
- Investigated elevated root activity
- Correlated suspicious outbound network activity

7. MITRE ATT&CK Mapping

Mapped attacker behavior to official MITRE ATT&CK techniques including:

- T1059.004 — Unix Shell
- T1071 — Application Layer Protocol
- T1548 — Abuse Elevation Control Mechanism
- T1053.003 — Cron
- T1098 — Account Manipulation
- T1543 — Create or Modify System Process

Key Findings

- Socat-based reverse shell activity detected
- Outbound TCP communication identified
- Successful root privilege escalation confirmed
- Cron persistence mechanisms discovered
- SSH authorized_keys persistence identified
- Suspicious systemd service creation detected
- Multiple attacker shell execution events correlated

Files Included

- Linux_Incident_Response_and_Threat_Hunting_Investigation_Report.pdf
- MITRE_ATTACK_Mapping_Report.pdf
- Investigation screenshots
- Investigation command/query logs

Real-World Relevance

This project reflects real-world SOC analyst and incident response workflows used in enterprise environments. The investigation demonstrates how Linux forensic analysis and Auditd logging can be leveraged to identify compromise indicators, reconstruct attacker behavior, and improve threat detection visibility.

The project emphasizes practical blue team skills relevant to:

- SOC Analyst roles
- Incident Response roles
- Threat Hunting roles
- Linux Security Monitoring
- Security Operations